

Answers:

1. *Why is it important to be able to convert vectors to angles in game development?*
 - a. Generally, being able to convert to angles can be important for doing things like determining an object's orientation or implementing things like steering behaviours in a simple way.
2. *Explain the terms magnitude and direction in the context of vectors.*
 - a. Magnitude refers to the length or size of a vector, and is often denoted as $\|\vec{v}\|$. Direction describes the orientation of the vector.
3. *What are polar coordinates, and how do they differ from Cartesian coordinates?*
 - a. Polar coordinates (r, θ) represent points in space using their distance "r" from the origin, and an angle θ with respect to the positive x-axis. Cartesian coordinates define points based on their horizontal and vertical distance from the origin.
4. *Convert (3, 4) from Cartesian coordinates to polar coordinates.*
 - a. (5, 0.93 radians)
5. *Convert (-1, 3) from Cartesian coordinates to polar coordinates.*
 - a. (3.16, 1.89 radians)
6. *Convert (3, -2) from Cartesian coordinates to polar coordinates.*
 - a. (3.6, -0.59 radians)
7. *Convert $(1, \pi/2)$ from polar coordinates to Cartesian coordinates.*
 - a. (0, 1)
8. *Convert (1, 0) from polar coordinates to Cartesian coordinates.*
 - a. (1, 0)
9. *Convert $(5, \pi/4)$ from polar coordinates to Cartesian coordinates.*
 - a. (3.53, 3.53)
10. *Why do acos and asin fail to determine angles uniquely?*
 - a. Recall that $\cos(\theta)$ can return the same value for 2 different angles. An easy example is $\cos(\pi/4)$ and $\cos(-\pi/4)$, and thus cannot uniquely identify an angle in the range $[0, 2\pi]$. The input to acos is restricted to work in the range $[0, \pi]$. The function asin has a similar limitation, for example, $\sin(\pi/4)$ $\sin(3\pi/4)$ are the same, since asin is restricted to work in the range $[-\frac{\pi}{2}, \frac{\pi}{2}]$.
11. *What function would you normally use instead?*
 - a. atan2
12. *Describe how storing a player's orientation in polar coordinates may be advantageous over storing it as a vector in Cartesian coordinates.*

- a. Storing the player's orientation in polar coordinates may simplify rotations, as you can operate on the angle θ directly. Turning left or right is as simple as adding or subtracting from θ . Additionally, smoothly blending between orientations may be a lot easier when working in polar coordinates.